ReadMe: cluster_GPs_cleaning.ipynb

Step 1 — Load NFHS GPS Clusters (Both Rounds): GPS cluster shapefiles are loaded for NFHS4 and NFHS5. State names are standardized to uppercase before filtering (NFHS4 uses title case; NFHS5 uses uppercase). Zero-coordinate clusters are dropped. Both datasets are filtered to Rajasthan and Uttar Pradesh and reprojected to UTM Zone 44N (EPSG:32644) for accurate distance calculations in metres. NFHS4 yields 5,266 clusters (3,849 rural) and NFHS5 yields 4,843 clusters (3,824 rural) in RJ+UP.

Step 2 — Load SHRUG Village Polygons: Village boundary polygons for Rajasthan (44,976 villages) and Uttar Pradesh (107,709 villages, split across two files due to GitHub's 100MB file size limit) are loaded and concatenated into a single GeoDataFrame of 152,685 villages, reprojected to UTM Zone 44N.

Step 3 — Load LGD Crosswalk and Assign GP Codes to Villages: The SHRUG-to-LGD crosswalk links each SHRUG village to its governing Gram Panchayat via the Ministry of Panchayati Raj's Local Government Directory. A composite `shrid2`-format key is constructed from each village's Census 2011 state, district, subdistrict, and village codes and used to join GP LGD codes onto the village polygons. Villages converted to Urban Local Bodies are handled using the retained `old_gp_lgd_code` field. GP coverage is 87.6% for Rajasthan villages and 52.0% for UP villages, with the lower UP rate reflecting a known ~40% LGD data coverage gap concentrated in eastern UP and Bundelkhand.

Step 4 — Buffer Analysis: Raw GP Coverage: For each cluster, a displacement-radius buffer is created — 5km for rural clusters and 2km for urban clusters, per the DHS GPS displacement protocol. A vectorised spatial join (`gpd.sjoin`) identifies all village polygons with valid GP codes that intersect each buffer, and unique GPs per cluster are counted. This step produces preliminary results before the district constraint is applied.

Step 5 — Nearest Village Join and District-Constrained GP Classification: Each cluster is assigned to its nearest SHRUG village polygon via `gpd.sjoin_nearest`, providing the cluster's Census district code (`pc11_d_id`). Per the DHS GPS displacement README, cluster displacement is restricted to stay within the same admin2 (district) area, making cross-district GP misassignment impossible. The buffer GP list is therefore filtered to retain only GPs whose villages fall within the same district as the cluster's nearest matched village. Clusters are then classified as:

- **Clean**: Only one GP within the district-constrained buffer. Treatment assignment is unambiguous regardless of displacement direction.
- **At-risk**: Two or more GPs within the district-constrained buffer. GP assignment could change depending on displacement direction.
- **No GP match**: No GP found within the district-constrained buffer — primarily due to the UP LGD coverage gap or peri-urban Out Growth areas.

Saving Outputs: Cluster-to-GP crosswalk CSVs are saved for each survey round to `../data/`, containing cluster identifiers, GPS coordinates, matched village and GP codes, and classification flags.